

Decision Trees

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can fit a root question and leaf predictions from training examples.

I can trace a new example through the learned tree.

I can compare a shallow tree with a baseline and state its limits.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

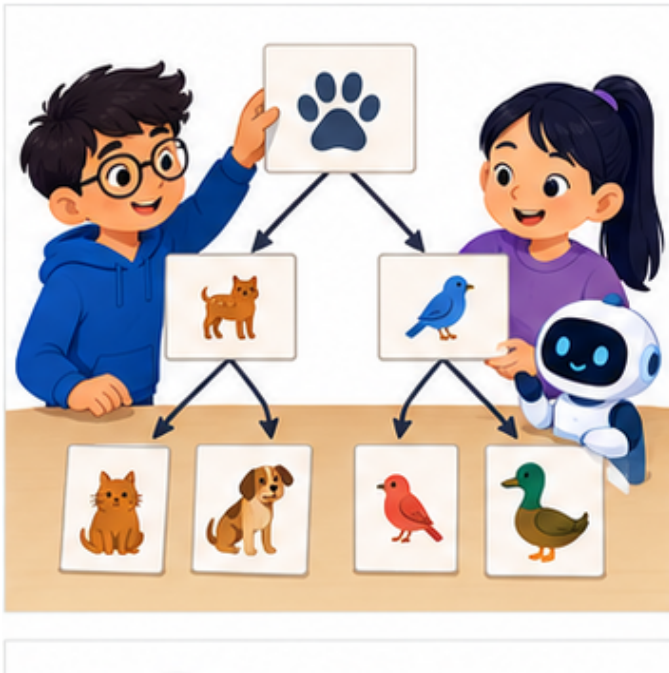
Without opening the old lesson, explain one key idea from Training and Testing. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 14: Decision Trees

Each question partitions possibilities; question order affects efficiency.



THINKER MISSION

Build a tree that identifies eight mystery animals.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Why does a cutoff with zero training errors still need validation on separate examples?

This week's end goal

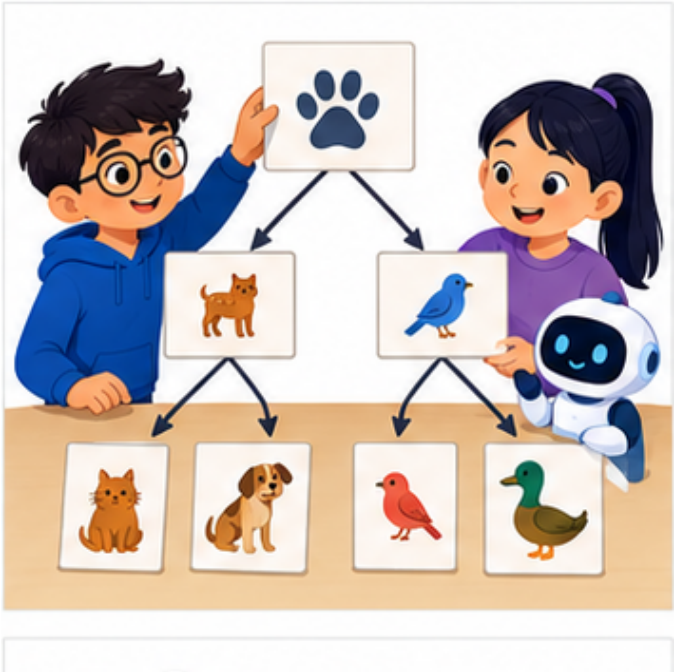
By the end, I can fit a one-question tree using training labels and an explicit candidate comparison.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use decision trees to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A decision tree asks a feature question and follows a branch to a prediction. A tree with one question has depth one and is sometimes called a stump. Unlike Week 8, where the two output labels were fixed in advance, this week we fit both the question and the label stored at each leaf.

Our six fictional training rows have two binary sensor flags available before prediction: near and tall. A value of 1 means that flag is present; 0 means absent. The reviewed scene label is separate. Each row represents a different trip on a familiar route, including rows with identical values. Similar readings can have different scene labels. These are teaching examples, not robot control instructions.

Row · Near · Tall · Label

Row: 1

Near: 1

Tall: 0

Label: blocked

Row: 2

Near: 1

Tall: 1

Label: blocked

Row: 3

Near: 1

Tall: 0

Label: blocked

Row: 4

Near: 0

Tall: 1

Label: clear

Row: 5

Near: 0

Tall: 0

Label: clear

Row: 6

Near: 0

Tall: 1

Label: blocked

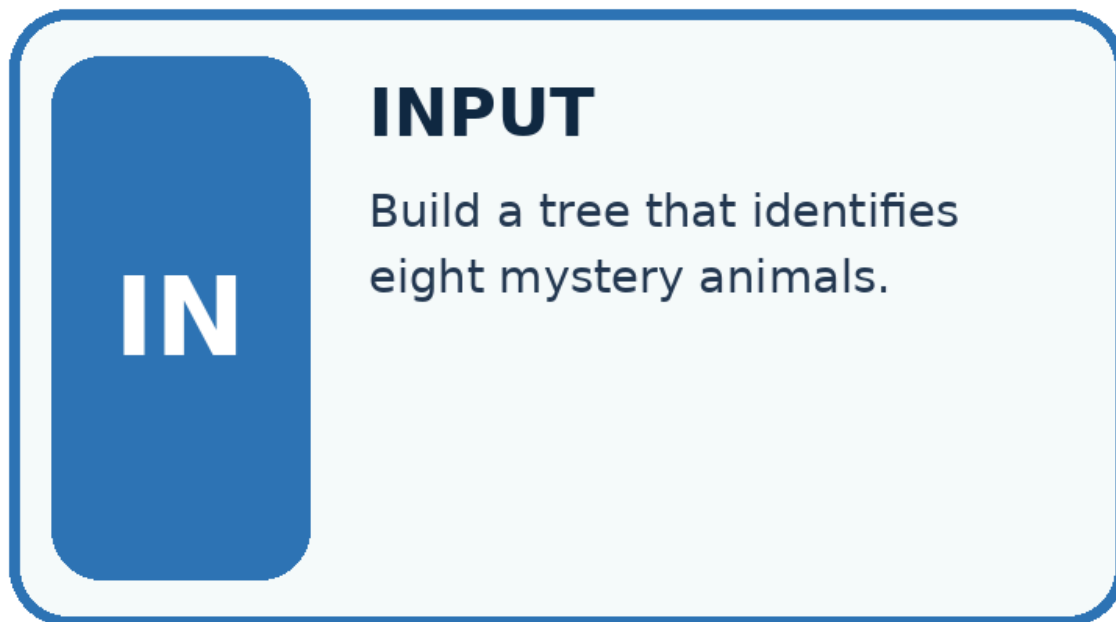
Try near first and tall second. For each question, group the training rows by 0 and 1, then predict each group's majority label; choose clear for tied label counts. Count all training mistakes and keep the first question if scores tie. Stop after one question. This is our stated classroom fitting procedure.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



PROCESS

Each question partitions possibilities; question order affects efficiency.



OUT

OUTPUT

Build a tree that identifies eight mystery animals.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
rows = [(1, 0, "blocked"), (1, 1, "blocked"),
        (1, 0, "blocked"), (0, 1, "clear"),
        (0, 0, "clear"), (0, 1, "blocked")]

def majority(labels):
    if labels.count("blocked") > labels.count("clear"):
        return "blocked"
    return "clear"

best_errors = len(rows) + 1

for feature in [0, 1]:
    leaves = {}
    for value in [0, 1]:
        labels = [row[2] for row in rows
                  if row[feature] == value]
        leaves[value] = majority(labels)

    errors = 0
    for row in rows:
        if leaves[row[feature]] != row[2]:
            errors = errors + 1

    print(feature, errors)

    if errors < best_errors:
        best_errors = errors
        tree = (feature, leaves)

print(tree[0], tree[1][0], tree[1][1])
```

Feature 0 is near; feature 1 is tall; row[2] is the label. Output is 0 1, then 1 2, then 0 clear blocked. Near makes one training mistake: near=0 contains two clear labels and one blocked, so that leaf predicts clear. Near=1 contains three blocked labels. Tall makes two mistakes because both its groups have a blocked majority.

The stored tree contains the chosen feature and its two learned leaves. The code assumes nonempty data, binary flags, allowed labels and examples on both branches of each candidate. An empty branch would need a declared fallback; this small example does not implement one. The score is total training mistakes, not an average giving small and large branches equal weight.

This is an explicitly limited depth-one learner. General tree algorithms can repeat split selection within branches and often use impurity criteria such as Gini. Our code does not implement that full procedure. Keeping the scope small lets you inspect every decision.

Connect the picture and the explanation

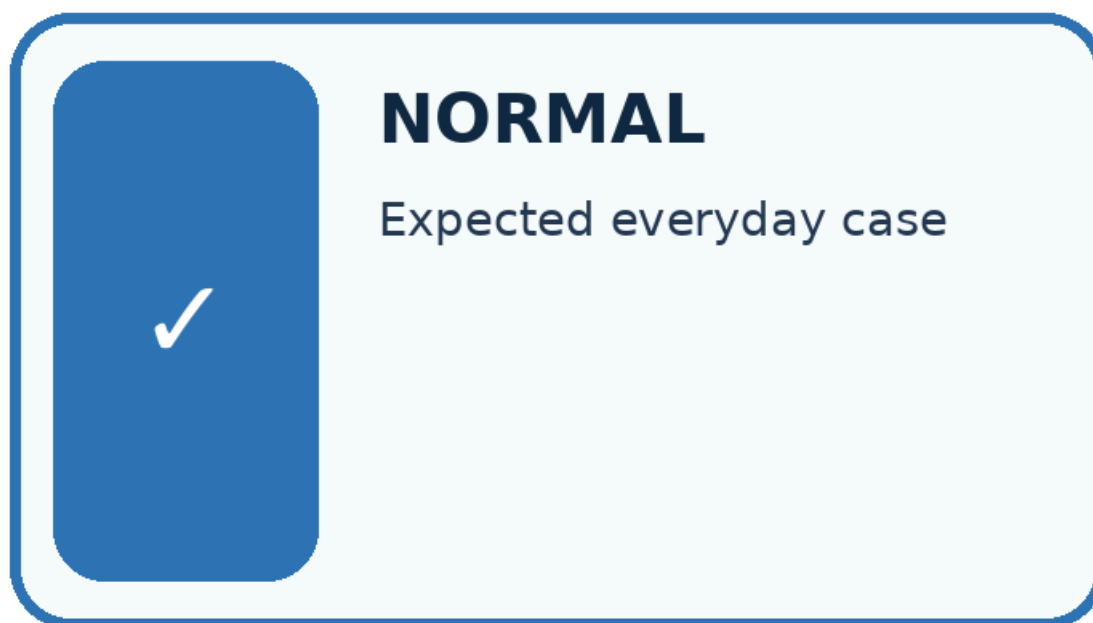
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





BOUNDARY

Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: New input becomes (0,1).

Prediction: The fitted tree predicts Yes.

Reason: It follows diagram=1; cover colour is not the selected question.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

DECISION TREE

questions connected by
branches

2

NODE

a question or result point

3

LEAF

final output

4

PARTITION

divide possibilities into groups

Word · My own example · Not this / boundary

Word: decision tree

My own example: A sequence of branching questions that routes an input to an output at a leaf.

Not this / boundary: _____

Word: node

My own example: A point in a graph or tree representing a state, question or result.

Not this / boundary: _____

Word: leaf

My own example: A final node of a decision tree that supplies an output rather than another question.

Not this / boundary: _____

Word: partition

My own example: One separated portion of a dataset, such as training, validation or test data.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Root

Meaning: The first question asked by a tree

Word: Branch

Meaning: A path selected by the answer to a question

Word: Depth

Meaning: The largest number of question steps from root to leaf

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 14 • WORKED EVIDENCE

- 1 Start with 8 animals.
- 2 Question A leaves groups of 4 and 4; question B leaves 7 and 1.
- 3 A balanced first split usually reduces later questions.
- 4 Trace every animal to verify one and only one leaf.

1. Start with 8 animals.
2. Question A leaves groups of 4 and 4; question B leaves 7 and 1.
3. A balanced first split usually reduces later questions.
4. Trace every animal to verify one and only one leaf.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Two paths through the learned tree

Root question · Branch answer · Learned leaf

Root question: Is near equal to 1

Branch answer: No near is 0

Learned leaf: Predict clear

Root question: Is near equal to 1

Branch answer: Yes near is 1

Learned leaf: Predict blocked

Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

Validation rows are (near=1,tall=1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear). Follow only near: predictions are blocked, clear, clear, blocked. Two of four match. The training-majority baseline predicts blocked, since training has four blocked and two clear; it also matches two of four validation labels.

Which validation rows does the tree miss, and which does the baseline miss? Explain why the tall value cannot change this tree's prediction.

Keep the fitted root and leaves unchanged while scoring. Equal accuracy does not mean identical errors, and five correct training predictions out of six is not the validation result.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Decision Trees

A fictional toy classifier predicts whether a book needs an illustration label.

The facts we will use

Training rows (colour_cover,digram_inside,label): (1,1,Yes),(1,0,No),(0,1,Yes),(0,0,No).

Candidates ask colour_cover or diigram_inside; each leaf uses its training majority. A leaf tie chooses No. Select fewer training errors.

Decision Trees

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Colour candidate

Both colour leaves tie Yes/No, choose No: two training errors.



2. Diagram candidate

Diagram 1 leaf predicts Yes; diagram 0 leaf predicts No: zero errors.



3. Select tree

Choose the diagram question using the training comparison.



4. Apply new input

A new (1,0) row follows diagram=0 and predicts No.

CHANGE ONE THING

New input becomes (0,1).

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Colour candidate

What happens and why: Both colour leaves tie Yes/No, choose No: two training errors.

Step: 2. Diagram candidate

What happens and why: Diagram 1 leaf predicts Yes; diagram 0 leaf predicts No: zero errors.

Step: 3. Select tree

What happens and why: Choose the diagram question using the training comparison.

Step: 4. Apply new input

What happens and why: A new (1,0) row follows diagram=0 and predicts No.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A question becomes a fitted tree only after its leaves and selection procedure are specified. The training comparison prefers the feature that separates these labels. Prediction follows the selected question; it does not reselect the tree using a new case's answer. Save candidate leaves and errors so the choice can be repeated.

What this example does not establish

Zero errors on four toy examples do not establish that diagrams always determine a real book's classification.

Read the labelled picture: Why does changing colour alone not change this tree's prediction?

Change one thing: New input becomes (0,1). Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: A decision tree learns because arrows are drawn between boxes.
Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Record the two candidate questions, branch label counts, total errors, tie rules, depth limit, stored tree and validation predictions. Retain all rows, including the blocked example in the near=0 branch; removing a disagreement would change the evidence.

Our tree uses a single feature because that question scored better under this procedure. This neither establishes that near causes blockage nor that tall is useless in every dataset. Next week a nearest-neighbour method will predict by comparing examples instead of following learned questions.

A learner says the tree is reliable because near=1 has only blocked training labels. Rewrite this claim using the leaf evidence, validation score and one limitation.

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Do not assume near=1 must always predict blocked. If a new training set gives that branch a clear majority, this same learner will store clear there. A feature name is not the answer. Use the training labels belonging to that branch.

Adding more questions can fit finer training details but does not guarantee better new-case performance. Record any depth choice made during development, and keep final test data separate. A leaf majority describes this training group; it is not a promise about every future example.

Practise with less help: Verify the candidate scores

Use workbook W2 for the connected example “Fitting a decision tree”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Fitting a decision tree. Proposed explanation: “A near branch must predict blocked”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why does changing colour alone not change this tree’s prediction?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Choose yes/no questions, build a tree for eight cards, count questions per card, and exchange trees for an unseen trace test.

Choose yes/no questions, build a tree for eight cards, count questions per card, and exchange trees for an unseen trace test.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

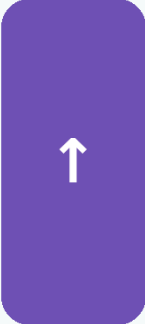
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER

1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Create a transparent tree for hard constraints before applying preference scores.

THIS WEEK'S CHECKPOINT Create a transparent tree for hard constraints before applying preference scores.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Fit or hand-specify recommender classification rules transparently, retaining candidate definitions and training-only selection evidence.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Build a tree that identifies eight mystery animals.

CHALLENGE Build a tree that identifies eight mystery animals.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 14 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

First ask animal book? If no, Other. If yes, ask pages<100; yes gives Short Animal, no gives Longer Animal.

The question

Trace 90-page animal and science books.

Worked explanation

The animal book reaches Short Animal; science reaches Other after the first question.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

A decision tree can stop early

Evidence or stage · Value or result

Evidence or stage: Not an animal book

Value or result: Other

Evidence or stage: Animal and fewer than 100 pages

Value or result: Short Animal

Evidence or stage: Animal and at least 100 pages

Value or result: Longer Animal

First ask animal book? If no, Other. If yes, ask pages<100; yes gives Short Animal, no gives Longer Animal.

Trace 90-page animal and science books.

The animal book reaches Short Animal; science reaches Other after the first question.

14. Try a new case independently

Independent case: Fit a new tree independently

New training rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree. Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree predictions and compare accuracy with the training-majority baseline.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 14 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A tree asks topic is space, then pages at most 60. A space book has 70 pages. Trace both answers and explain why you cannot stop at the first question.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Create a transparent tree for hard constraints before applying preference scores.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Build a tree for science books: first ask science?; then, for yes, pages $\leq$ 100? Trace science/100 and history/70. What should happen to science with missing pages?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional toy classifier predicts whether a book needs an illustration label.

By the end, I can · How I will show it

By the end, I can: Fit a root question and leaf predictions from training examples

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Trace a new example through the learned tree

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Compare a shallow tree with a baseline and state its limits

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Fit a root question and leaf predictions from training examples

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional toy classifier predicts whether a book needs an illustration label.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? A new (1,0) row follows diagram=0 and predicts No.

1. Select tree
2. Diagram candidate
3. Colour candidate
4. Apply new input

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Apply new input A new (1,0) row follows diagram=0 and predicts No. A question becomes a fitted tree only after its leaves and selection procedure are specified. The training comparison prefers the feature that separates these labels. Prediction follows the selected question; it does not reselect the tree using a new case's answer. Save candidate leaves and errors so the choice can be repeated.

Practice 2: explain the deciding evidence

Changed-case practice: New input becomes (0,1). Which conclusion is supported by the supplied facts?

1. The fitted tree predicts Yes.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The fitted tree predicts Yes. It follows diagram=1; cover colour is not the selected question.

Repair a missing building block

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and prediction pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?. Return to this week and explain what you changed.

Review Training and Testing (ai-2/11). Goal: Choose a split unit that matches the intended prediction claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan counts?. Return to this week and explain what you changed.

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit evidence and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.